

Milestone: (2)

Benchmark (Results and Analysis):

1. Throughput Analysis:

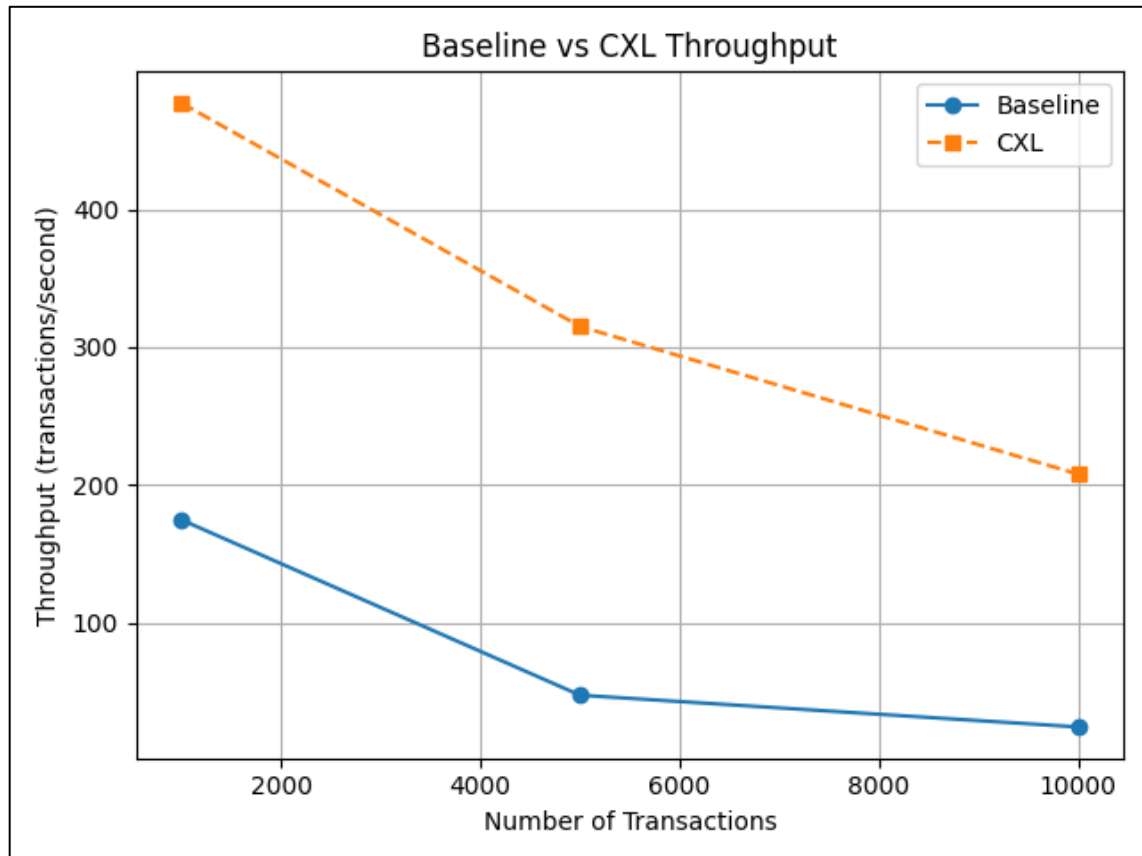


Figure 1: Baseline vs CXL Throughput

The results demonstrate that CXL technology achieved higher throughput compared to the baseline system across all tested workload sizes. At a workload of 1,000 transactions, CXL throughput reached approximately 477.30 transaction/second, compared to 174.69 transactions/second for the baseline system. At 5,000 transactions, CXL achieved a rate of approximately 315.03 transactions/second, versus 47.27 transactions/second for the baseline; at 10,000 transactions, CXL throughput reached approximately 207.65 transactions/second, compared to 24.17 transactions/second for the baseline.

Overall, the baseline system's throughput dropped significantly as the number of transactions increased, whereas CXL maintained a higher throughput rate, indicating superior performance in handling larger workloads within this prototype.

2. Validation Latency Analysis:

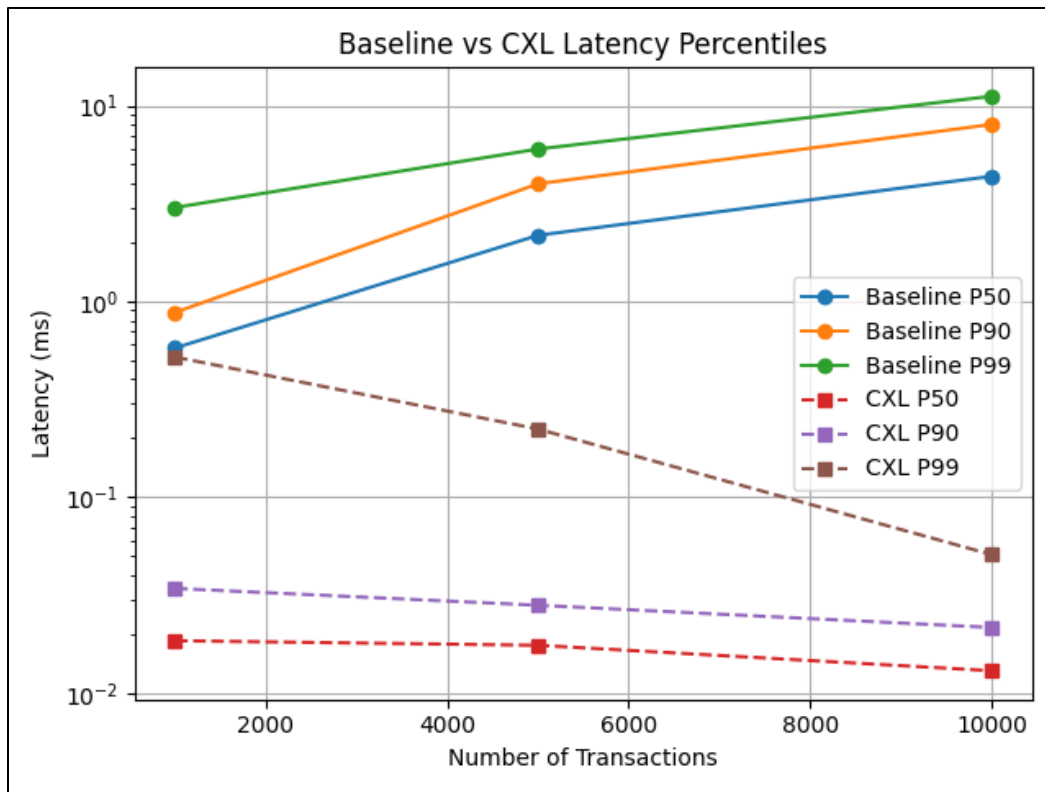


Figure 2: Baseline vs CXL Percentiles

The results show that baseline validation latency increases with the workload size. At a level of 10,000 transactions, baseline values were approximately 4.58 ms for the 50th percentile (p50), 8.34 ms for the 90th percentile (p90), and 12.40 ms for the 99th percentile (p99).

In contrast, CXL values were significantly lower, measuring approximately 0.013 ms for the 50th percentile (p50), 0.022 ms for the 90th percentile (p90), and 0.051 ms for the 99th percentile (p99). This indicates a marked reduction in validation overhead when using the CXL-aware validation mechanism implemented in the prototype.

3. Cumulative Distribution Function (CDF) Analysis:

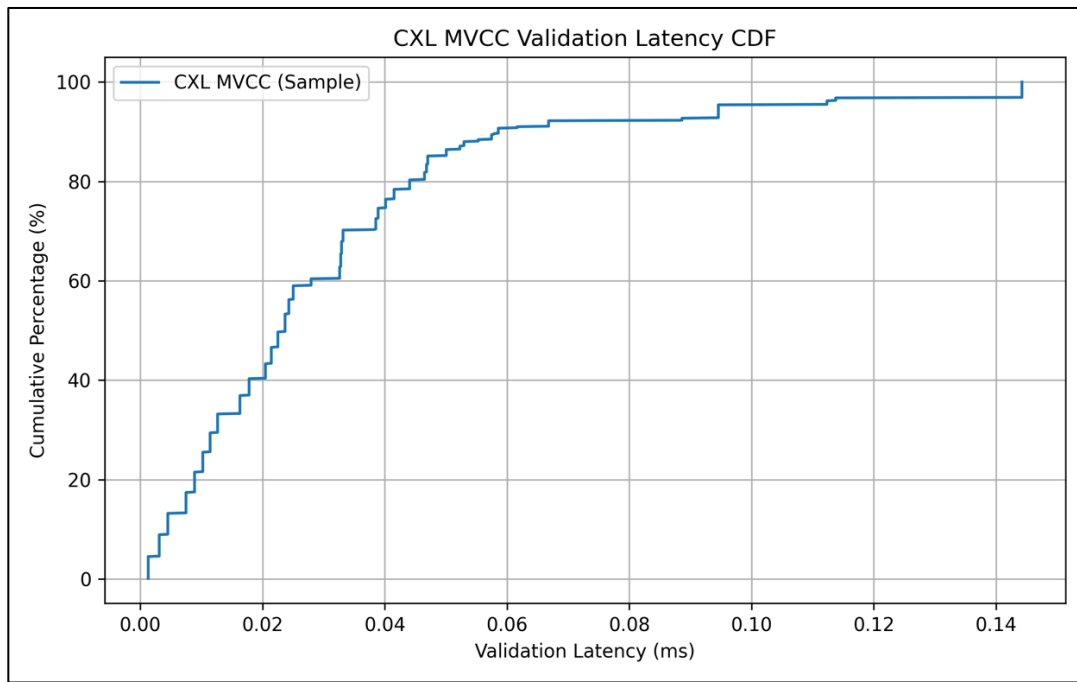


Figure 3: CXL MVCC Validation Latency CDF

The Cumulative Distribution Function (CDF) illustrates the distribution of the validation process response time (latency), based on a sample of 1,000 execution traces.

The results indicate that most validation operations are completed quickly, with a limited number of high values appearing at the tail of the distribution; the CDF helps clarify the response time distribution and reveals the "tail latency" behavior of the validation process.

4. Resource Utilization Analysis:

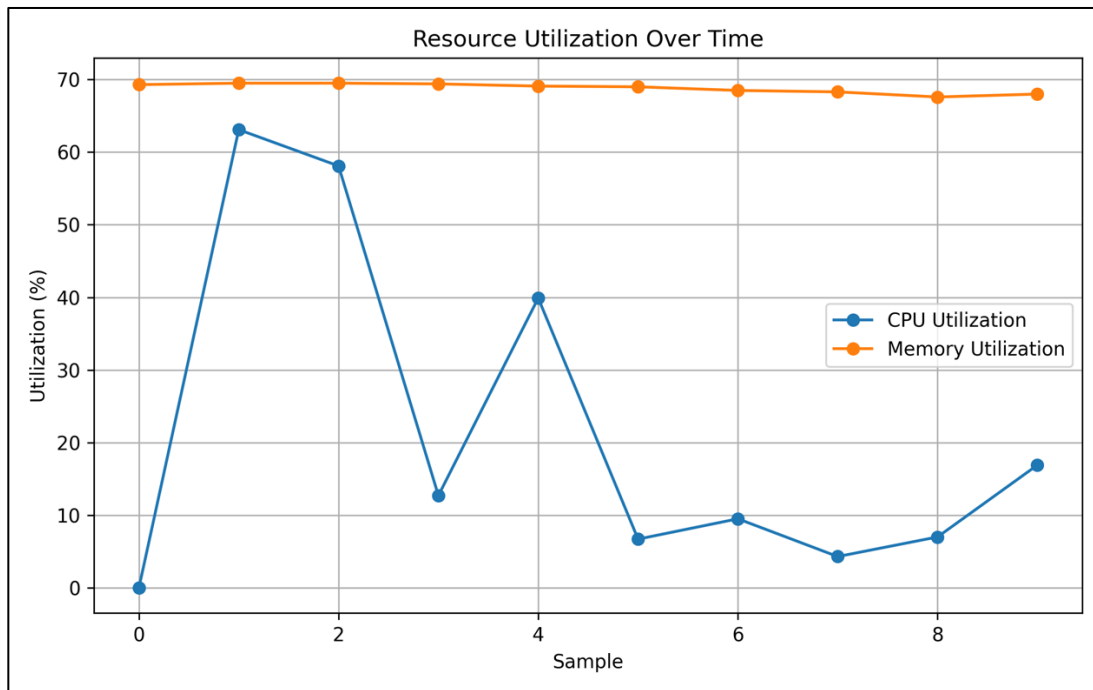


Figure 4: Resource Utilization Over Time

The resource utilization analysis shows that memory usage remained relatively stable throughout the measurement period, whereas CPU utilization experienced significant fluctuations, exceeding 60% in some samples.

This indicates that the workload causes variations in CPU utilization during process execution, while memory usage remained more stable during the monitoring period.

5. Overall Performance Analysis:

Overall, the prototype results demonstrated that the CXL-aware MVCC validation mechanism achieved higher throughput and lower validation latency compared to the baseline system (including the current test environment); resource utilization results also showed relatively stable memory usage, despite some variation in CPU utilization.

However, it should be noted that the experiment was conducted using a simulated environment for the prototype rather than actual CXL hardware; therefore, the results cannot be generalized to all database systems. Future work may involve testing the system using real CXL hardware, larger-scale workloads, and more diverse scenarios.